12 A light wave of amplitude A is incident normally on a surface of area S. The intensity reaching the surface is I.

The amplitude of the light wave is increased to $2A$. The light is then focused on to a smaller area $\frac{1}{3}S$.

What is the intensity on this smaller area?

A $6I$ **B** $12I$ **C** $18I$ **D** $24I$